

Citation addendum to eleven notes of *“Structured Sectors of the Collatz Carry Equation: Christoffel Towers, 2-Adic Windows, and Shell Residents”*

Elias De Jesús

Independent Researcher

ORCID: 0009-0007-0190-9143

dejesuselias10@gmail.com

September 2026

Purpose

Eleven notes in this collection use the 2-adic Sturmian carry constant Ξ_α , or the first-disagreement bit depth $\ell(D) = v_2(Q(D) + \Xi_\alpha)$ built from it, and none of them cites López and Stoll. This addendum supplies that citation for all eleven at once. It accompanies them without altering any of them.

A separate addendum, *Citation addendum and correction to “The 2-Adic Sturmian Carry Constant in the Collatz Carry Equation”* (September 2026), covers the note in which Ξ_α is introduced; this one covers the notes that consume it.

No theorem, proof or numerical result in any of the eleven notes changes, and this addendum makes no new mathematical claim.

1 The omitted citation, and what it contains

J. López and P. Stoll, *The $3x+1$ conjugacy map over a Sturmian word*, *Integers* **9** (2009), #A13, 141–162 [2].

For every irrational slope $\alpha \in (0, 1)$ with convergents p_k/q_k , and with $1c_\alpha(j) = \lceil (j+1)\alpha \rceil - \lceil j\alpha \rceil$, their Theorem 1 gives, in $\mathbb{Z}_2$,

$$\Phi(1c_\alpha) = -\frac{1}{3} - \sum_{j \geq 0} (-1)^{j+1} \frac{2^{q_{j+1}+q_j-1}}{3(3^{p_{j+1}} - 2^{q_{j+1}})(3^{p_j} - 2^{q_j})}, \quad (1)$$

where Φ is the Bernstein–Lagarias conjugacy map [1]; their Corollary 2 gives a generalized continued fraction for $-1/\Phi(1c_\alpha)$, convergent in $\mathbb{Z}_2$. Their §4 leaves $\mathbb{Z}_2$ for the real function $\Phi_{\mathbb{R}}$, proving irrationality of $\Phi_{\mathbb{R}}(m_\alpha)$ on $\mathbb{Q} \cap (\ln 2 / \ln 3, 1]$ and of a dual limit point below $\ln 2 / \ln 3$; that restriction is forced, since $\sum 2^{nq}/3^{np}$ converges in $\mathbb{R}$ exactly when $\ln 2 / \ln 3 < p/q \leq 1$. The companion paper [3] proves aperiodicity for parity words of ones-density strictly above $\ln 2 / \ln 3$.

Why this bears on the eleven notes. The denominators of (1) are the walls of two consecutive convergent shells and its term exponents are $q_{j+1} + q_j - 1$, which in the collection’s coordinates read $p_n + p_{n+1} - 1$. *These are the approximants and the approximation depths that the bit analyzer measures.* The constant is theirs, up to sign and normalization:

$$\Xi_\alpha = -\Phi(1c_{\ln 2 / \ln 3}),$$

their slope being the ones-density $\ln 2 / \ln 3 = 1/\alpha$ and their convergent (p_k, q_k) the collection’s shell (q_n, p_n) with numerator and denominator exchanged. Verified modulo 2^{3000} .

2 Attribution, as it should read in each of the eleven

One sentence suffices, at the point where Ξ_α or $\ell(D)$ is first introduced:

The constant Ξ_α and the convergent-shell approximation depths it measures are not new here. For every irrational slope, López and Stoll [2] give a closed 2-adic series for the Bernstein–Lagarias conjugacy value of the characteristic Sturmian word, whose term exponents are $p_n + p_{n+1} - 1$ in the present coordinates; $\Xi_\alpha = -\Phi(1c_{\ln 2 / \ln 3})$. The companion note’s Theorem 7.1 is an independent derivation, sharper in separating the upper $(p_n - 1)$ and lower $(p_n + p_{n+1} - 1)$ convergent cases with exact constants, shell by shell.

Two further points travel with it, both from the companion addendum:

- **Range.** The companion note’s Theorem 7.1 holds as proved for $n \geq 3$ in that note’s own shell numbering, where $(q_1, p_1) = (1, 1)$; its proof uses $q_n \geq 2$ and $q_{n+1} > q_n$, which first hold at the shell $(2, 3)$. Any note quoting the per-shell depth inherits that range.
- **Partial correspondence.** Only the odd-indexed continued-fraction convergents of [2] are periodic cycle values; the even ones carry an additional gap term.

3 The eleven notes, and where the attribution belongs

Each line quotes the note’s own text at the point of first use, with its page.

note	quoted line	p.
A Bit Analyzer for the Collatz Carry Equation	“Definition 2.1 (Bit analyzer). The bit depth of D is $\ell(D) = v_2(Q(D) + \Xi_\alpha)$ ”	2
Adjacent Tower Products in the Collatz Carry Equation	“analyzer is $\ell(D) = v_2(Q(D) + \Xi_\alpha)$, where Ξ_α is the 2-adic Sturmian carry constant of the second note”	1
Finite Christoffel Products in the Collatz Carry Equation	“bit depth $\ell(D) = v_2(Q(D) + \Xi_\alpha)$ with Ξ_α the archived Sturmian...”	2
Standard-Block Nonsingularity in the Collatz Carry Equation	“the 2-adic Sturmian carry constant of the companion note pins the only...”	1
Grouped Tower Products in the Collatz Carry Equation	“ $v_2(Q_{a,b} + \Xi_\alpha) = 83 = p_5 + p_6 - 1$ ”	1
Tail-Tracking and the Leading-Block Principle	“The bit depth $\ell(D) = v_2(Q(D) + \Xi_\alpha)$ is governed by a...”	1
The Wrap Regime in the Collatz Carry Equation	“algorithm computes $\ell(D) = v_2(Q(D) + \Xi_\alpha)$ for every finite product of standard tower blocks”	1
The Positive-Entropy 2-Adic Carry Law	“Let Ξ_α denote the rigid Sturmian 2-adic carry constant associated to...”	8
Unbalanced Residents: the (7, 11) Shell	“so $\ell(D_{-17}) = v_2(Q + \Xi_\alpha) = 2$ ”	3
Unbalanced Residents and Time-Axis Contacts	“has computable $\ell(D) = v_2(Q(D) + \Xi_\alpha)$ ”	2
The Anatomy of the -17 Cycle	“ $\ell(D_{-17}) = v_2(Q + \Xi_\alpha) = 2$ ”	5

Scope. The remaining sixteen notes of the collection contain no occurrence of Ξ_α and no first-disagreement depth, and need no attribution; nor does the referee note `cubic_modulus_threshold_referee_note.md`, whose subject is the modulus depth $\theta = m/S$, a different quantity. Together with the note in which Ξ_α is introduced, that partitions the collection’s 29 files as $1 + 11 + 16 + 1$.

Acknowledgment. The omission was identified during an adversarial citation audit of the Sturmian thread, assisted by Claude (Anthropic). All statements here, and any errors, are the author’s.

References

- [1] Daniel J. Bernstein and Jeffrey C. Lagarias. The $3x+1$ Conjugacy Map. *Canadian Journal of Mathematics*, 48(6):1154–1169, 1996.
- [2] Josefina López and Peter Stoll. The $3x+1$ Conjugacy Map over a Sturmian Word. *Integers*, 9(A13):141–162, 2009.
- [3] Josefina López and Peter Stoll. The $3x+1$ Periodicity Conjecture in $\mathbb{R}$, 2021.